STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 26

REGRESSION ANALYSIS

SIMPLE LINEAR REGRESSION

PREPARED BY
HAZBER SAMSON
FAST-NUCES ISLAMABAD

REGRESSION ANALYSIS

INTRODUCION: At the beginning of the nineteenth century, **Legendre and Gauss** published papers on the *method of least squares*, which implemented the earliest form of what is now known as *linear regression*. The term *regression* was first introduced by **Francis Galton**. In a famous paper, Galton found that, although there was a tendency for tall parents to have tall children and for short parents to have short children, the average height of children born of parents of a given height tended to move or "regress" toward the average height in the population as a whole.1 In other words, the height of the children of unusually tall or unusually short parents tends to move toward the average height of the population. Galton's *law of universal regression* was confirmed by his friend Karl Pearson, who collected more than a thousand records of heights of members of family groups.2 He found that the average height of sons of a group of tall fathers was less than their fathers' height and the average height of sons of a group of short fathers was greater than their fathers' height, thus "regressing" tall and short sons alike toward the average height of all men. In the words of Galton, this was "regression to mediocrity. The modern interpretations of regression is, as follows however,

DEFINITION: Regression analysis is a statistical technique for investigating and modeling the relationship between variables.

Applications of regression are numerous and occur in almost every field, including engineering, the physical and chemical sciences, economics, management, life and biological sciences, and the social sciences. In fact, regression analysis may be the most widely used statistical technique.

Usually, Researchers want more than just finding the best linear approximation of one variable given a set of others. They want valid statistical relationships. They want to draw conclusions about what happens if one of the variables actually changes. That is: they want to say something about things that are not observed (yet). In this case, they want to find a fundamental relationship. To do this it is assumed that there is a general relationship that is valid for all possible observations from a well-defined population. Restricting attention to linear relationships, we specify a statistical model as linear regression model. Some examples of analyses using regression models include the following:

• Estimating weight gain by the addition to children's diet of different amounts of various dietary supplements
• Predicting scholastic success (grade point ratio) based on students' scores on an aptitude or entrance test
• Estimating amounts of sales associated with levels of expenditures for various types of advertising
• Predicting fuel consumption for home heating based on daily temperatures and other weather factors
• Estimating changes in interest rates associated with the amount of deficit spending

Regression analysis is used to predict the value of one variable on the basis of the other variables. Regression analysis is generally classified into two kinds, **simple** and **multiple**. The regression will be **simple** if it involves only one independent variable and it will be **multiple** if it involves more than one independent variables.

SIMPLE LINEAR REGRESSION MODEL (SLRM)

Regression analysis is used to predict the value of one variable on the basis of the other variables. Regression analysis is generally classified into two kinds, simple and multiple. The regression will be simple if it involves only one independent variable and it will be multiple if it involves more than one independent variables.

By linear regression model we mean an approximate linear association between dependent and independent variables. There could be two types of linearity

1. Linearity in Parameters
2. Linearity in Variables

In case of SLRM both linearity's must hold.

If there is only one independent variable and one dependent variable than regression is called Simple regression. In this case regression equation forms a straight line so it is also called SLRM. It is also called the *two-variable linear regression model* or *bivariate linear regression model* because it relates the two variables *x* and *y*.
It is defined as

$$\boxed{Y = \alpha + \beta X + \varepsilon}$$

Where $Y =$ dependent variable
And $X =$ independent variable
Also $\alpha =$ regression coefficient or y- Intercept
$\beta =$ Regression coefficient or slope
$\varepsilon =$ Error term, Random term or Residual Term

Note that 1. $\alpha \, and \, \beta$ are also called parameters of the Model.
2. Different Names of dependent and independent variables are given as

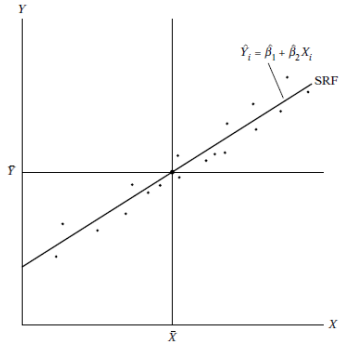
y	*x*
Dependent Variable	Independent Variable
Explained Variable	Explanatory Variable
Response Variable	Control Variable
Predicted Variable	Predictor Variable
Regressand	Regressor

Note that in $Y = \alpha + \beta X + \varepsilon$
$Y =$ Total variation
$\alpha + \beta X =$ Explained variation or Systematic variation, deterministic component of Y
$\varepsilon =$ Unexplained variation or Random Variation, stochastic component of Y
For example in demand function
$D = a + bP + f(income, taste, advertisement, other\ prices, Taxes\ etc)$ or $D = a + bP + \varepsilon$
Where $\varepsilon = Error\ Term = f(income, taste, advertisement, other\ prices, Taxes\ etc)$

HAZBER SAMSON

PROPERTIES OF REGRESSION LINE

1. Regression line always passes through the sample means $\overline{X}$ *and* $\overline{Y}$.



2. Mean value of the estimated values of Y is equal to the mean value of actual Y ie $\overline{\hat{Y}_i} = \overline{Y_i}$
3. The mean value of the residuals is zero ie $\overline{\varepsilon_i} = 0$
4. The residuals are uncorrelated with predicted values.
5. The residuals are uncorrelated with independent variables.

ESTIMATION OF REGRESSION LINE

We will discuss three methods for estimating the parameters α *and* β of the model
The Method of Least Squares
The Method of Moments
The Method of Maximum Likelihood

ORDINARY LEAST SQUARES (OLS)

OLS is technique for fitting the best straight line to the sample x, y observations. It involves minimizing the sum of squared (vertical) deviations of points from the line.
ie $Min \sum \left(Y - \hat{Y}\right)^2$

Note that Y : Observed Values
$\hat{Y}$: estimated values
If we define $Y - \hat{Y}_i = e_i$ than we have to $Min \sum e_i^2$

ASSUMPTIONS OF OLS

1. There exists linearity in parameters.
2. Error terms have zero mean
3. *No Heteroscedasticity*
   Error terms have constant variance for all observations.
4. *No Autocorrelation*
   The error terms corresponding to different observations are uncorrelated.
5. *No Multicollinearity*
   There is no exact relationship exists among two or more independent variables.

HAZBER SAMSON

ASSUMPTIONS OF CLASSICAL LINEAR REGRESSION MODEL

The Simple Linear Regression Model (SLRM), Gaussian or Standard is the cornerstone of most econometric theory. It has the following assumptions. Classical linear regression model (CLRM) is given by

$$Y_i = \alpha + \beta X_i + \varepsilon_i$$

ASSUMPTIONS OF SIMPLE LINEAR REGRESSION MODEL

ASSUMPTION-1 LINEAR REGRESSION MODEL *The regression model is linear in the parameters,* though it may or may not be linear in the variables. It means that the dependent variable can be calculated as a linear function of X, plus a disturbance term.

ASSUMPTION-2 NUMBER OF OBSERVATIONS The number of observations *n* must be greater than the number of parameters to be estimated: Alternatively, the number of observations must be greater than the number of explanatory variables.

ASSUMPTION-3 INDEPENDENT VARIABLE X HAS SOME VARIATION
By this assumption we mean that not all observations of X are the same; at least one has to be different so that the sample Var(X) is not 0. Furthermore, there should be **no outliers** in the values of the *X* variable.

ASSUMPTION-4 EXPECTED VALUE OF THE ERROR TERM IS ZERO $E(\varepsilon_i) = 0 \;\; for\, all\, i.$
Mean value of error terms is zero. ie This means that the disturbance is a genuine disturbance, so if we took a large number of samples the mean disturbance would be zero.

ASSUMPTION-5 HOMOSCEDASTICITY OF ERROR TERMS $Var(\varepsilon_i) = \sigma^2 \;\; for\, all\, i.$
Variance of error terms is same for all observations. This means that all error terms have the same variance regardless of the value of X.

ASSUMPTION-6 INDEPENDENCE OF ERROR TERMS $Cov(\varepsilon_i, \varepsilon_j) = 0 \;\; for\, all\, i \neq j$
Error terms are independent of each other. So there is no correlation between two error terms. In short, the observations are sampled independently.

ASSUMPTION-7 FIXED VALUES OF INDEPENDENT VARIABLES $Cov(X_i, \varepsilon_i) = 0 \;\; for\, all\, i\,.$
Values taken by the regressor *X* may be considered fixed in repeated samples also X variable(s) and the error term are independent of each other.

ASSUMPTION-8 NORMALITY OF ERROR TERMS $\varepsilon_i \sim N(0, \sigma^2) \;\forall\; i$
Error terms are normally distributed with mean zero and variance σ^2 .

HAZBER SAMSON

COEFFICIENT OF DETERMINATION

The coefficient of determination is statistic used to determine how well a regression is fit. It is the ratio of the explained variation to the total variation. It shows proportion of variation in y variable due to explanatory variables of the model. We can say that how much of the variability in y variable can be explained by the linear relationship of y to the independent variable x. It is denoted by R^2 *or* r^2 and is given by

$$R^2 = \frac{Explained\,Variation}{Total\,Variation} = \frac{SSR}{SST} = \frac{\sum\left(\hat{Y}-\bar{Y}\right)^2}{\sum\left(Y-\bar{Y}\right)^2}$$

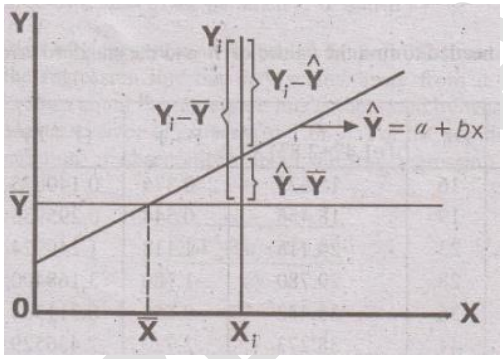
or

$$R^2 = 1 - \frac{Un\exp lained\,Variation}{Total\,Variation} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum\left(Y-\hat{Y}\right)^2}{\sum\left(Y-\bar{Y}\right)^2}$$

Note that Total Variation = Explained Variation + Unexplained Variation

$$i.e.\ SST = SSR + SSE \ \ or\ \ \sum(Y-\bar{Y})^2 = \sum(\hat{Y}-\bar{Y})^2 + \sum\left(Y-\hat{Y}\right)^2$$

Graphically



Also Note that

1. $Total\ Varation = SST = Sum\ of\ \ Squares\ Total = \sum(Y-\bar{Y})^2 = \sum Y^2 - \frac{\left(\sum Y\right)^2}{n}$
2. $SSR = Sum\ of\ Squares\ \text{Re}\ gression = \sum(\hat{Y}-\bar{Y})^2 = \sum(Y-\bar{Y})^2 - \sum\left(Y-\hat{Y}\right)^2$
3. $SSE = Sum\ of\ \ Squares\,Errors = \sum\left(Y-\hat{Y}\right)^2 = \sum Y^2 - a\sum Y - b\sum XY$

PROPERTIES OF COEFFICIENT OF DETERMINATION

1. It is a non-negative measure.
2. Its value lies between o and 1 i.e. $0 \le R^2 \le 1$
3. Coefficient of determination is just the square of Correlation coefficient. $R^2 = r^2$

INERPRETATION OF COEFFICIENT OF DETERMINATION

If $R^2 = 0.90$ than it means that 90% of the variation in dependent variable is due to the linear relationship of the independent variables where the remaining 10% of the variation is due other factors.

HAZBER SAMSON

ORDINARY LEAST SQUARES (OLS)

$$Y = \alpha + \beta X + \varepsilon$$

NORMAL EQUATIONS

- $\sum Y = n\alpha + \beta \sum X$
- $\sum XY = \alpha \sum X + \beta \sum X^2$

ESTIMATORS

Now

- $\hat{\beta} = \dfrac{n\left(\sum xy\right) - \left(\sum x\right)\left(\sum y\right)}{n\left(\sum x^2\right) - \left(\sum x\right)^2}$
- $\hat{\beta} = \dfrac{\sum \left(X - \overline{X}\right)\left(Y - \overline{Y}\right)}{\sum \left(X - \overline{X}\right)^2}$
- $\hat{\beta} = \dfrac{S_{XY}}{S_{XX}}$

Also

$$\hat{\alpha} = \overline{Y} - \hat{\beta}\overline{X}$$

STANDARD ERROR OF ESTIMATE or RESIDUAL STANDARD ERROR

RSE is a measure of the lack of fit of the model to the data

$$RSE = S_{y.x} = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{\sum \left(Y - \hat{Y}\right)^2}{n-2}} = \sqrt{\frac{\sum Y^2 - \alpha \sum Y - \beta \sum XY}{n-2}}$$

CO-EFFICIENT OF DETERMINATION

$$R^2 = \frac{Explained\ Variation}{Total\ Variation} = \frac{SSR}{SST} = \frac{\sum \left(\hat{Y} - \overline{Y}\right)^2}{\sum \left(Y - \overline{Y}\right)^2}$$

or

$$R^2 = \frac{Unexplained\ Variation}{Total\ Variation} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum \left(Y - \hat{Y}\right)^2}{\sum \left(Y - \overline{Y}\right)^2}$$

HAZBER SAMSON

TESTING OF PARAMETERS OF THE MODEL

TESTING OF INTERCEPT α

1. $H_0 : \alpha = 0$

$H_1 : \alpha \neq 0$

2. *Level of significance*

$\alpha = 0.05$

3. *Test Statistic*

$$t = \frac{\hat{\alpha} - \alpha}{S.E(\hat{\alpha})} \qquad \text{Where } S.E(\hat{\alpha}) = \sqrt{\frac{\sigma_e{}^2 \sum X^2}{n.\sum (X - \overline{X})^2}} \text{ also } \sigma_e{}^2 = \frac{\sum e^2}{n-2}$$

4. *Critical Region*

$|t_{cal}| \geq t_{\frac{\alpha}{2}(n-2)}$

5. If t_{cal} lies in CR than reject Ho otherwise do not reject H_0.

P-VALUE APPROACH $P = 2 \times P(X \geq |t_0|)$ *or* $P = 2 \times P([\Phi(|t_0|)])$

CONFIDENCE INTERVAL APPROACH

The $(1-\alpha)$ 100% for confidence interval for $\hat{\alpha}$ is

$$\hat{\alpha} - t_{\frac{\alpha}{2}(n-2)} \times S.E(\hat{\alpha}) \leq \alpha \leq \hat{\alpha} + t_{\frac{\alpha}{2}(n-2)} \times S.E(\hat{\alpha})$$

TESTING OF SLOPE β

1. $H_0 : \beta = 0$

$H_1 : \beta \neq 0$

2. *Level of significance*

$\alpha = 0.05$

3. *Test Statistic*

$$t = \frac{\hat{\beta} - \beta}{S.E(\hat{\beta})} \qquad \text{Where } S.E(\hat{\beta}) = \sqrt{\frac{\sigma_e{}^2}{\sum (X - \overline{X})^2}} \text{ also } \sigma_e{}^2 = \frac{\sum e^2}{n-2}$$

4. *Critical Region*

$|t_{cal}| \geq t_{\frac{\alpha}{2}(n-2)}$

5. If t_{cal} lies in CR than reject Ho otherwise do not reject H_0.

P-VALUE APPROACH $P = 2 \times P(X \geq |t_0|)$ *or* $P = 2 \times P([\Phi(|t_0|)])$

CONFIDENCE INTERVAL APPROACH

The $(1-\alpha)$ 100% for confidence interval for β is

$$\hat{\beta} - t_{\frac{\alpha}{2}(n-2)} \times S.E(\hat{\beta}) \leq \beta \leq \hat{\beta} + t_{\frac{\alpha}{2}(n-2)} \times S.E(\hat{\beta})$$

HAZBER SAMSON

EXAMPLES OF SIMPLE LINEAR REGRESSION MODEL

EXAMPLE consider the data given below

X	5	7	12	16	20
Y	4	12	18	21	24

(a) Find a regression equation of Y on X and interpret the Results.
(b) Find the value of Y when X = 25 using Fitted regression Model.
(c) In the regression equation $Y = \alpha + \beta X + \varepsilon$ check the significance of β
(d) In the regression equation $Y = \alpha + \beta X + \varepsilon$ check the significance of α

SOLUTION Using the scatter Plot in correlation analysis for this data we have observed that there exists a negative correlation between x and y.

(A) *FITTING THE MODEL* $Y = \alpha + \beta X + \varepsilon$

For calculations we shall consider only $Y = \alpha + \beta X$

We know that $\hat{\beta} = \dfrac{n\left(\sum XY\right) - \left(\sum X\right)\left(\sum Y\right)}{n\left(\sum X^2\right) - \left(\sum X\right)^2}$ *and* $\hat{\alpha} = \overline{Y} - \hat{\beta}\overline{X}$

Let's make table for the values used in formulas

X	Y	XY	X^2	Y^2	$\hat{Y}$	$Y - \hat{Y}$	$\left(Y - \hat{Y}\right)^2$	$Y - \overline{Y}$	$(Y - \overline{Y})^2$
5	4	20	25	16	7.26	-3.26	10.63	-11.8	139.24
7	12	84	49	144	9.7	2.3	5.29	-3.8	14.44
12	18	216	144	324	15.8	2.2	4.84	2.2	4.84
16	21	336	256	441	20.68	0.32	0.10	5.2	27.04
20	24	480	400	576	25.56	-1.56	2.43	8.2	67.24
60	**79**	**1136**	**874**	**1501**	**79**	**0**	**23.29**	**0**	**252.8**

Now $\hat{\beta} = \dfrac{n\left(\sum XY\right) - \left(\sum X\right)\left(\sum Y\right)}{n\left(\sum X^2\right) - \left(\sum X\right)^2} = \dfrac{(5)(1136) - (60)(79)}{(5)(874) - (60)^2} = \dfrac{940}{770} = 1.22$

Also $\overline{X} = \dfrac{\sum X}{n} = \dfrac{60}{5} = 12$ and $\overline{Y} = \dfrac{\sum Y}{n} = \dfrac{79}{5} = 15.8$

So $\hat{\alpha} = \overline{Y} - \hat{\beta}\overline{X} = 15.8 - (1.22)(12) = 1.16$

So the required regression equation will be given as $\hat{Y} = 1.16 + 1.22\,X$

INTERPRETATION OF RESULTS OF THE MODEL

Interpretation of $\hat{\alpha} = 1.16$**:** If X = 0 then Y will be 1.16.

Interpretation of $\hat{\beta} = 1.22$**:** If X is increased by 1 unit Y will be increased by 1.22 units on average in the long run keeping other factors as constant.

HAZBER SAMSON

(B) *VALUE OF Y FOR GIVEN X* Fitted regression equation is $\hat{Y} = 1.16 + 1.22\,X$

At $X = 25$ we have $Y = 1.16 + 1.22(25) = 31.66$

(C) *TESTING OF SLOPE*

CRITICAL-VALUE APPROACH

1. $H_0 : \beta = 0$

$H_1 : \beta \neq 0$

2. *Level of significance*

$\alpha = 0.05$

3. *Test Statistic*

$$t = \frac{\hat{\beta} - \beta}{S.E(\hat{\beta})} \quad \text{Where } S.E(\hat{\beta}) = \sqrt{\frac{\sigma_e^{\,2}}{\sum (X - \overline{X})^2}} \text{ also } \sigma_e^{\,2} = \frac{\sum e^2}{n-2}$$

X	Y	XY	X^2	$\hat{Y}$	$Y - \hat{Y}$	$(Y - \hat{Y})^2$	$X - \overline{X}$	$(X - \overline{X})^2$
5	4	20	25	7.26	-3.26	10.63	-7	49
7	12	84	49	9.7	2.3	5.29	-5	25
12	18	216	144	15.8	2.2	4.84	0	0
16	21	336	256	20.68	0.32	0.10	4	16
20	24	480	400	25.56	-1.56	2.43	8	64
60	**79**	**1136**	**874**	**79**	**0**	**23.29**	**0**	**154**

Now $\sigma_e^{\,2} = \frac{\sum e^2}{n-2} = \frac{23.29}{5-2} = 7.76$ and $S.E(\hat{\beta}) = \sqrt{\frac{7.76}{154}} = 0.2245$ So $t = \frac{1.22 - 0}{0.2245} = 5.43$

4. *Critical Region*

$|t_{cal}| \geq t_{\frac{\alpha}{2}(n-2)} = t_{0.025(3)} = 3.182$

5. Since t_{cal} lies in Critical Region So we reject H_0 and conclude that β is significant.

P-VALUE APPROACH $P = 2 \times P([\Phi(|t_0|) = 2 \times P([\Phi(|3.182|) = 0.025$

Since $p - value < \alpha$

$\Rightarrow$ *reject* $H_0 \Rightarrow \beta$ *is significant.*

CONFIDENCE INTERVAL APPROACH

The $(1-\alpha)$ 100% for confidence interval for β is

$$\hat{\beta} - t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\beta}) \leq \beta \leq \hat{\beta} + t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\beta})$$

$$(1.22) - (3.182)(0.2245) \leq \beta \leq (1.22) - (3.182)(0.2245)$$

$$0.5056 \leq \beta \leq 1.9344$$

Since 0 *does not lie in confidence interval*

$\Rightarrow$ *reject* $H_0 \Rightarrow \beta$ *is significant.*

HAZBER SAMSON

(D) *TESTING OF INTERCEPT*

CRITICAL-VALUE APPROACH

1. $H_0 : \alpha = 0$

$H_1 : \alpha \neq 0$

2. *Level of significance*

$\alpha = 0.05$

3. *Test Statistic*

$$t = \frac{\hat{\alpha} - \alpha}{S.E(\hat{\alpha})} \qquad \text{Where } S.E(\hat{\alpha}) = \sqrt{\frac{\sigma_e{}^2 \sum X^2}{n.\sum (X - \overline{X})^2}} \text{ also } \sigma_e{}^2 = \frac{\sum e^2}{n-2}$$

X	Y	XY	X^2	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$	$X-\overline{X}$	$(X-\overline{X})^2$
5	4	20	25	7.26	-3.26	10.63	-7	49
7	12	84	49	9.7	2.3	5.29	-5	25
12	18	216	144	15.8	2.2	4.84	0	0
16	21	336	256	20.68	0.32	0.10	4	16
20	24	480	400	25.56	-1.56	2.43	8	64
60	**79**	**1136**	**874**	**79**	**0**	**23.29**	**0**	**154**

Now $\sigma_e{}^2 = \dfrac{\sum e^2}{n-2} = \dfrac{23.29}{5-2} = 7.76$

And $S.E(\hat{\alpha}) = \sqrt{\dfrac{\sigma_e{}^2 \sum X^2}{n.\sum (X-\overline{X})^2}} = \sqrt{\dfrac{7.76 \times 874}{5 \times 154}} = 2.9678$

So $t = \dfrac{\hat{\alpha} - \alpha}{S.E(\hat{\alpha})} = \dfrac{1.16 - 0}{2.9678} = 0.3909$

4. *Critical Region*

$$|t_{cal}| \geq t_{\frac{\alpha}{2}(n-2)} \Rightarrow |t_{cal}| \geq t_{\frac{0.05}{2}(5-2)} \Rightarrow |t_{cal}| \geq t_{0.025(3)} \Rightarrow |t_{cal}| \geq 3.182$$

5. Since t_{cal} does not lie in Critical Region So we do not reject H_0 and conclude that α is insignificant.

P-VALUE APPROACH $P = 2 \times P([\Phi(|0.3909|) = 2 \times P([\Phi(|3.182|) = 2 \times 0.358 = 0.716$

Since $p-value > \alpha \Rightarrow$ *do no reject* $H_0 \Rightarrow \alpha$ *is insignificant.*

CONFIDENCE INTERVAL APPROACH

The $(1-\alpha)$ 100% for confidence interval for α is

$$\hat{\alpha} - t_{\frac{\alpha}{2}(n-2)} \times S.E(\hat{\alpha}) \leq \alpha \leq \hat{\alpha} + t_{\frac{\alpha}{2}(n-2)} \times S.E(\hat{\alpha})$$

$$(1.16) - (3.182)(2.9678) \leq \alpha \leq (1.16) + (3.182)(2.9678)$$

$$-8.28 \leq \alpha \leq 10.60$$

Since 0 *lies in confidence* int *erval* $\Rightarrow$ *do not reject* $H_0 \Rightarrow \alpha$ *is insignificant.*

HAZBER SAMSON